Lecture 27

Compact orbits and geometric quantisation

Objectives. *A compact coadjoint orbit has a symplectic form. Not every such orbit supplies a representation of the chosen global group. We calculate the obstruction on a sphere. The symplectic area of the sphere divided by $2\pi\hbar$ must be integral. An equivariant lift imposes a further condition when the group is* $\mathrm{SO}(3)$. *Holomorphic sections then connect the orbit to the representations of the previous lecture.*

By the end of this lecture you will be able to:

- Compute the coadjoint-orbit form of $\mathrm{SU}(2)$ with its radius normalisation.
- Define a unitary connection locally and prove the sphere integrality condition.
- Distinguish an ordinary prequantum bundle from an equivariant one.
- Recover spin $m/2$ by holomorphic sections and keep this construction separate from shifted character formulas.

27.1 The orbit sphere and its area

Use the basis of Lecture 4,

$$X_j = -\frac{i}{2}\sigma_j, \qquad [X_1, X_2] = X_3$$

with cyclic versions of the bracket. The Pauli matrices satisfy

$$\sigma_i\sigma_j = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k.$$

Identify $\mathfrak{su}(2)^*$ with $\mathbb{R}^3$ by $x_j = \ell(X_j)$. For $u = (u_1, u_2, u_3)$ put $X_u = \sum_j u_j X_j$. Then $[X_u, X_v] = X_{u\times v}$.

The adjoint action is the rotation cover from Lecture 4. Its dual action is the same rotation action in these orthonormal coordinates. Thus its orbits are the spheres

$$S_r^2 = \{x : |x| = r\} \quad (r > 0)$$

and the origin. The infinitesimal action is $\xi_u(x) = u \times x$. Indeed, pairing it with v gives $(u \times x)\cdot v = -x \cdot (u \times v)$, which is exactly the dual derivative from Lecture 19.

Proposition 27.1. *Let $r > 0$. The orbit form at $x \in S^2_r$ is*

$$\omega_x(v, w) = \frac{x \cdot (v \times w)}{r^2}, \qquad v, w \in T_x S^2_r.$$

If Ω is the outward-oriented area form of the sphere, then $\omega = \Omega/r$. In spherical coordinates,

$$\omega = r \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2_r} \omega = 4\pi r. \tag{27.1}$$

Proof. The KKS formula of Lecture 19 requires

$$\omega_x(u \times x, v \times x) = x \cdot (u \times v).$$

At $x = re_3$, the two tangent vectors are $r(u_2, -u_1, 0)$ and $r(v_2, -v_1, 0)$. Substitution in the stated formula gives $r(u_1 v_2 - u_2 v_1)$, the required value. Rotation invariance gives the formula at every x.

The outward area form is $\Omega_x(v, w) = x \cdot (v \times w)/r$. Thus $\omega = \Omega/r$. The usual sphere parametrisation has area form $r^2 \sin\theta \, d\theta \wedge d\phi$, giving the spherical formula (27.1). Integration over $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$ gives $4\pi r$. ∎

Write $\xi_j = \xi_{e_j}$ for the standard basis vectors e_j. At the north pole, $\xi_1 = -r\partial_{x_2}$ and $\xi_2 = r\partial_{x_1}$, and $\omega(\xi_1, \xi_2) = r$. This is a convenient sign check. The origin must be treated separately: it is a point orbit, and none of the formulas dividing by r applies there.

Fix a positive constant $\hbar$. It sets the scale at which we compare symplectic area with phases. Keeping it explicit will make the integrality condition easier to track.

27.2 Connections described in local frames

A Hermitian line bundle is a complex line bundle with a smoothly varying positive inner product on its fibres. A local *unitary frame* is a section of norm one that never vanishes. In such a frame, a section is described by a complex coefficient function u.

Definition 27.2. A *unitary connection* differentiates local coefficients by

$$\nabla u = du + Au,$$

where A is an imaginary-valued 1-form. Let unitary frames e_i on U_i and e_j on U_j have connection forms A_i and A_j. If $e_j = g_{ij} e_i$, with $g_{ij} : U_i \cap U_j \to \mathrm{U}(1)$, the connection forms are required to satisfy

$$A_j = A_i + g_{ij}^{-1} dg_{ij}.$$

The *curvature* F_∇ is the imaginary-valued 2-form given on U_i by $F_\nabla = dA_i$.

Here du is the ordinary differential, evaluated on tangent vectors. A section s has coefficients u_i and u_j in the two frames. Since $e_j = g_{ij} e_i$, these coefficients satisfy $u_i = g_{ij} u_j$, or equivalently $u_j = g_{ij}^{-1} u_i$. Differentiating $u_i = g_{ij} u_j$ gives

$$du_i + A_i u_i = g_{ij}\big(du_j + (A_i + g_{ij}^{-1} dg_{ij}) u_j\big).$$

The left side is the coefficient of ∇s in the frame e_i. In the frame e_j this coefficient is $du_j + A_j u_j$, and coefficients change by the factor g_{ij}. Hence frame independence of ∇s for every section is exactly the rule for A_j in the definition.

Since g_{ij} is scalar, $d(g_{ij}^{-1} dg_{ij}) = 0$ locally. Indeed, write $g_{ij} = e^{i\alpha}$ with real α, so this form is $i\,d\alpha$. Hence $dA_i = dA_j$ and the curvature is global.

The same connection can be written in any nonvanishing frame. The change-of-frame formulas remain valid for $g_{ij} \in \mathbb{C}^\times$, with complex-valued A_i. Only unitary frames require imaginary-valued forms.

Along a path, the equation for parallel transport is $du + Au = 0$. In one frame its solution multiplies u by $\exp(-\int A)$. This is a phase because A is imaginary. Curvature measures the resulting phase around small loops. In physics language, A is a gauge potential and ∇ is a covariant derivative.

Definition 27.3. A *prequantum line bundle* for (M, ω) at scale $\hbar$ is a Hermitian line bundle with unitary connection satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega. \tag{27.2}$$

The sign is part of our convention. A prequantum bundle is an initial geometric input. Its space of all smooth sections is usually far larger than the desired representation. A polarisation, such as holomorphicity, supplies a further selection.

We cite the general prequantisation theorem. A closed real 2-form ω on a smooth manifold admits a prequantum line bundle at scale $\hbar$ exactly when its periods divided by $2\pi\hbar$ are integers. A *period* of ω is the integral of ω over a closed oriented surface mapped smoothly into the manifold, or over an integer combination of such surfaces. Equivalently, $[\omega/(2\pi\hbar)]$ is an integral cohomology class. See Kirillov, Chapter 1, Section 2.4.

We prove the sphere case directly.

27.3 The integral sphere bundle

Put

$$m = \frac{2r}{\hbar}.$$

Take a northern chart excluding the south pole and a southern chart excluding the north pole. Define

$$A_N = -\frac{im}{2}(1 - \cos\theta)d\phi, \qquad A_S = \frac{im}{2}(1 + \cos\theta)d\phi. \tag{27.3}$$

Although $d\phi$ is singular at each pole, A_N is smooth at the north pole and A_S at the south pole. To see this we use a complex coordinate, and we recall its rules. For $z = x + iy$ put

$$dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy, \qquad \partial_z = \tfrac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

Then $df = \partial_z f\, dz + \partial_{\bar{z}} f\, d\bar{z}$ for a function f, and $dz \wedge d\bar{z} = -2i\, dx \wedge dy$. Writing the 1-form $a\,dz + b\,d\bar{z}$ in dx and dy and using the rule of Lecture 19 gives

$$d(a\,dz + b\,d\bar{z}) = (\partial_z b - \partial_{\bar{z}} a)\, dz \wedge d\bar{z}.$$

In the northern complex coordinate $z = \tan(\theta/2)e^{i\phi}$,

$$A_N = \frac{m}{2}\frac{z\,d\bar{z} - \bar{z}\,dz}{1+|z|^2}.$$

This expression is smooth at $z = 0$. The southern formula has the same property in its own coordinate. In physics language, these are the potentials of a magnetic monopole on two patches.

Differentiation gives

$$dA_N = dA_S = -\frac{im}{2}\sin\theta\,d\theta\wedge d\phi = -\frac{i}{\hbar}\omega.$$

On the overlap, $A_S - A_N = im\,d\phi$. Thus A_N and A_S satisfy the change-of-frame formula if the frame transition is

$$e_S = e^{im\phi}e_N.$$

This is a single-valued function on the overlap exactly when $m \in \mathbb{Z}$. For positive radius this means m is a positive integer. In physics language, $m \in \mathbb{Z}$ is Dirac's quantisation condition.

The idea of the necessity proof below is one loop with two caps. Transport around a loop multiplies a fibre by a phase. In a frame on a disk D bounded by the loop, Stokes' theorem makes this phase $\exp(-\int_D F_\nabla)$. In the plane of Lecture 19, a loop bounds one disk, and the Weyl relation of Lecture 18 has a phase of the same kind, a constant times an area. No condition arises there. On a sphere, a loop bounds two disks, with opposite orientations, as Fig. 27.1 shows. The two disks must give the same phase. Hence $\exp(-\int_{S^2} F_\nabla) = 1$, which says that $\int_{S^2}\omega \in 2\pi\hbar\mathbb{Z}$. We now write this with frames. In physics language, the latitude phase of Exercise 27.2 is the Berry phase of a spin.



Figure 27.1: One latitude loop, oriented by increasing ϕ, on two copies of the sphere. Its arrow points the same way in both. In the first copy, the shaded north cap D_N has this loop as its oriented boundary. In the second copy, the shaded south cap D_S has the reversed loop as its oriented boundary, so its phase has the opposite sign in the exponent.

Proposition 27.4. *The sphere S^2_r, $r > 0$, admits a prequantum line bundle if and only if $2r/\hbar \in \mathbb{Z}$. For $m = 2r/\hbar$, the preceding construction supplies one.*

Proof. Sufficiency is the explicit construction. For necessity, suppose that S^2_r carries a prequantum line bundle.

Step 1 (unitary frames on the two disks). Use the two open disk charts obtained by deleting the opposite poles. On each chart, parallel transport a unit vector at its origin along radial

segments. Cover each radial segment by finitely many local unitary frames. The change-of-frame law makes transport agree on overlaps. The resulting finite composition of ODE solutions depends smoothly on the endpoint and gives a smooth unitary frame. Each chart contains a neighbourhood of its closed hemisphere, so these frames are valid up to and beyond the equator. Call them e_N and e_S, with connection forms A_N and A_S.

Step 2 (the transition function and its lift). Write $e_S = g e_N$ near the equator, with $g : S^1 \to \mathrm{U}(1)$. Orient the equator by increasing ϕ, as the boundary of the northern disk.

Lift $g(e^{i\phi})$ to a real phase $\alpha(\phi)$ for $0 \le \phi \le 2\pi$, using the circle covering from Lecture 4. Since the endpoints give the same value, $\alpha(2\pi) - \alpha(0) = 2\pi k$ for an integer k.

Step 3 (Stokes). By the change-of-frame law, $A_S = A_N + g^{-1}dg$. Moreover $g^{-1}dg = i\,d\alpha$, since $g = e^{i\alpha}$ along the equator. Let D_N and D_S be the closed northern and southern hemispheres. We apply the calculus Stokes formula for a 1-form on each oriented disk. The southern disk induces the opposite orientation on the equator. Hence

$$\int_{S^2} F_\nabla = \int_{D_N} dA_N + \int_{D_S} dA_S = \int_{S^1} (A_N - A_S) = -\int_{S^1} g^{-1}dg = -2\pi i k.$$

By the prequantum condition (27.2) and the spherical formula (27.1),

$$k = \frac{i}{2\pi}\int F_\nabla = \frac{1}{2\pi\hbar}\int \omega = \frac{2r}{\hbar}.$$

Hence $2r/\hbar \in \mathbb{Z}$. ∎

Since $m > 0$, the curvature has no global primitive on the sphere. If $F = dA$ globally, Stokes' theorem would give $\int_{S^2} F = 0$. The period is nonzero because $m \neq 0$. The two local potentials are therefore essential.

27.4 Holomorphic sections and equivariance

First we show that $O(m)$, with one metric, carries the prequantum connection. Its holomorphic sections are then the quantisation. Finally we show that the natural SU(2) action is the equivariant lift, and that SO(3) needs even m.

The metric and the connection

Identify the sphere with $\mathbb{CP}^1$ by

$$z \longmapsto \frac{r}{1+|z|^2}\big(2\operatorname{Re} z, 2\operatorname{Im} z, 1-|z|^2\big).$$

The north pole is $z = 0$, and $z = \tan(\theta/2)e^{i\phi}$ is the northern coordinate used above. Recall from Lecture 26 that SU(2) acts on lines by $g\cdot[v] = [gv]$. The identification map is equivariant for this action and the coadjoint action. Indeed, the line of $v = (1, z)$ corresponds to the traceless Hermitian matrix proportional to

$$vv^*/\|v\|^2 - I/2.$$

Conjugation by g sends this matrix to the matrix for the line of gv, and it rotates the Pauli coordinates of the matrix. With $|z| = \tan(\theta/2)$, we have $d|z| = \frac{1}{2}(1+|z|^2)\,d\theta$ and $r\sin\theta = 2r|z|/(1+|z|^2)$. Substitution in the spherical formula (27.1) gives the sphere form

$$\omega = \frac{2ir\,dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Since $i\,dz\wedge d\bar z = 2\,d(\operatorname{Re} z)\wedge d(\operatorname{Im} z)$, this form has the same positive orientation as the spherical formula (27.1).

The holomorphic frame transition of $O(m)$ in Lecture 26 is $e_S^{\mathrm{hol}} = z^m e_N^{\mathrm{hol}}$. The Hermitian metric with

$$\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-m}$$

gives unitary frames $e_N = (1+|z|^2)^{m/2}e_N^{\mathrm{hol}}$ and e_S whose transition is $(z/|z|)^m = e^{im\phi}$. This is the transition of the integral sphere bundle, so the potentials (27.3) define a unitary connection on $O(m)$.

The metric is SU(2)-invariant. To see this, we describe it fibre by fibre. A point f of the fibre over a line $[v]$ is a function on that line with $f(\lambda v) = \lambda^m f(v)$ for $\lambda\in\mathbb{C}^\times$. In this model, the holomorphic sections of Lecture 26 are the homogeneous polynomials of degree m, restricted to each line. On the fibre over $[v]$ put

$$\|f\|^2 = \frac{|f(v)|^2}{\|v\|^{2m}}.$$

Replacing v by λv multiplies the numerator and the denominator by $|\lambda|^{2m}$. Hence the norm does not depend on the chosen vector v. An element $g\in\mathrm{SU}(2)$ sends f to the point gf over $[gv]$ with $(gf)(gv) = f(v)$. Since g is unitary, $\|gv\| = \|v\|$, and so $\|gf\| = \|f\|$. The northern frame e_N^{hol} takes the value 1 at $v = (1,z)$, as the polynomial v_0^m does. Since $\|(1,z)\|^2 = 1+|z|^2$, this norm agrees with the metric displayed above.

A connection is *compatible* with the holomorphic structure when its form in a holomorphic frame has no $d\bar z$ term. Apply the change-of-frame law to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-m/2}e_N$ and use the coordinate formula for A_N above. The form of this connection in the frame e_N^{hol} is then

$$A_N^{\mathrm{hol}} = A_N - \frac{m}{2}\,d\log(1+|z|^2) = -m\frac{\bar z\,dz}{1+|z|^2}.$$

Exercise 27.4 carries out this frame change in coordinates. The form A_N^{hol} has only a dz term, and its curvature is

$$dA_N^{\mathrm{hol}} = m\frac{dz\wedge d\bar z}{(1+|z|^2)^2} = -\frac{i}{\hbar}\omega.$$

Thus the prequantum bundle can be taken to be $O(m)$, with its usual holomorphic structure.

For a given metric, the unitary connection compatible with the holomorphic structure is unique. Let e be a holomorphic frame and put $h = \|e\|^2$. Let A be the connection form in the frame e. The unitary frame $h^{-1/2}e$ has an imaginary-valued connection form A_u. The frame e is $h^{1/2}$ times this unitary frame, so the change-of-frame law gives $A = A_u + \frac{1}{2}d\log h$. Moreover $\bar A_u = -A_u$, and $\log h$ is real. Hence

$$A + \bar A = d\log h.$$

For a function f, write $\partial f = (\partial f/\partial z)\,dz$ and $\bar\partial f = (\partial f/\partial \bar z)\,d\bar z$, so that $df = \partial f + \bar\partial f$. Suppose that A has no $d\bar z$ term. Then $\bar A$ has no dz term. Comparing the dz terms on the two sides gives

$$A = \partial \log h.$$

Thus the metric determines the compatible unitary connection. For $h = (1+|z|^2)^{-m}$ this formula gives $-m\bar z\,dz/(1+|z|^2)$, which is A_N^{hol} again.

The quantisation

When $m = 2r/\hbar$ is a positive integer, the *holomorphic quantisation* of S^2_r is the section space

$$Q(S^2_r) = H^0(\mathbb{CP}^1, O(m)) \cong V_m.$$

By the section computation of Lecture 26, this space has dimension $m+1$ and spin $j = m/2 = r/\hbar$. An invariant positive inner product comes from compact averaging. Explicitly, the formula

$$\|s\|^2 = \frac{1}{4\pi r}\int_{S^2_r} \|s\|^2_{\mathrm{fibre}}\,\omega$$

is the norm of Exercise 26.4. Indeed, in the coordinate z, the section z^k of $O(m)$ has squared fibre norm $|z|^{2k}(1+|z|^2)^{-m}$, and with $u = |z|^2$ this integral is

$$\int_0^\infty u^k(1+u)^{-m-2}\,du = \frac{k!(m-k)!}{(m+1)!}.$$

The point orbit uses the trivial line and gives the trivial representation, without a radius denominator.

The equivariant lift

Definition 27.5. Let K be a Lie group with Lie algebra $\mathfrak{k}$, and let $M \subset \mathfrak{k}^*$ be a coadjoint orbit with its KKS form ω. An *equivariant prequantum bundle* for K is a prequantum line bundle for (M,ω) at scale $\hbar$ with a smooth K-action on its total space. The action must cover the coadjoint action on M, be linear and unitary on fibres, preserve the connection, and satisfy the *orbit normalisation*. This normalisation requires that, at each $\ell \in M$, the stabiliser K_ℓ act on the fibre over ℓ with differential

$$X \mapsto i\ell(X)/\hbar, \qquad X \in \mathrm{Lie}(K_\ell).$$

The orbit normalisation fixes the infinitesimal lift. Without it, merely asking for a bundle action would not specify its relation to the orbit functional.

At the north pole, the stabiliser in $\mathrm{SU}(2)$ is generated by $\exp(tX_3)$, of period 4π. In an equivariant prequantum bundle, this stabiliser acts on the fibre by the character

$$\exp(tX_3) \longmapsto e^{irt/\hbar} = e^{imt/2}.$$

This is a character exactly when $m \in \mathbb{Z}$. It is the action on the fibre of $O(m)$ from Lecture 26: $t_\theta = \exp(-2\theta X_3)$ acts there by $e^{-im\theta}$. The natural $\mathrm{SU}(2)$ action is linear on fibres and preserves the invariant metric. Each $g \in \mathrm{SU}(2)$ acts by a holomorphic bundle map. It sends

holomorphic frames to holomorphic frames. Its holomorphic coordinate change sends forms with only a dz term to forms of the same kind. Hence g transports the connection to a unitary connection compatible with the holomorphic structure. By uniqueness, the transported connection is the original one. Therefore the natural action supplies the required lift.

For SO(3) the same stabiliser rotation has period 2π. Its required character is single-valued only when $e^{2\pi i r/\hbar} = 1$, or m is even. Conversely, for even m the centre $-I$ acts trivially on every fibre of $O(m)$, so the action descends. Therefore

ordinary prequantum bundle	$m \in \mathbb{Z}$
SU(2)-equivariant prequantum bundle	$m \in \mathbb{Z}$
SO(3)-equivariant prequantum bundle	$m \in 2\mathbb{Z}$

for the positive-radius sphere, with $m > 0$. An ordinary integral sphere with odd m does exist. It does not quantise equivariantly for the global rotation group SO(3). Figure 27.2 shows the first integral spheres.

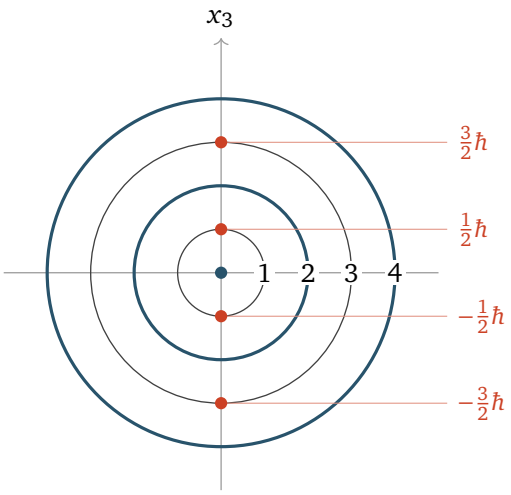


Figure 27.2: Cross-sections of the integral spheres, labelled by $m = 2r/\hbar$ with radius $r = \hbar m/2$, and the point orbit at the origin. Heavy circles have even m and admit an SO(3)-equivariant lift. The marks on the axis give the four heights $x_3 = \frac{\hbar}{2}k$, for $k = 3, 1, -1, -3$, associated with the weights of V_3, the quantisation of the sphere $m = 3$. Only the two extreme marks are points of that sphere on the axis.

Phase changes and the odd-degree obstruction

One might try to change the SU(2) lift by phases so that $-I$ acts trivially. We show that, for this sphere and its fixed connection, every such phase change is trivial.

Throughout this subsection, an action on $O(m)$ means a smooth action that is linear and unitary on fibres, as in the definition of an equivariant prequantum bundle. Suppose two such SU(2) actions cover the same sphere action and preserve the connection. For fixed g, compare their fibre maps. Their ratio is a bundle automorphism covering the identity. It is linear and unitary on each fibre, and each fibre is a complex line. Hence the ratio

is multiplication by a smooth function $a_g : S^2 \to \mathrm{U}(1)$. Since both actions preserve the connection,

$$a_g^{-1} da_g = 0.$$

The sphere is connected, so a_g is constant. Call this constant $\chi(g)$. The group-action law then implies

$$\chi(gh) = \chi(g)\chi(h).$$

Thus changing the lift while preserving the connection can only twist it by a continuous character of SU(2).

Every such character is trivial. Its differential is a real Lie algebra map from $\mathfrak{su}(2)$ to the abelian algebra $i\mathbb{R}$, so it vanishes on brackets. Since

$$[\mathfrak{su}(2), \mathfrak{su}(2)] = \mathfrak{su}(2),$$

the differential vanishes. Since the connected group SU(2) is generated by exponentials (Lecture 2), the character is identically one. Consequently a phase change of this kind cannot alter the action of the centre. Hence the natural action of Lecture 26 is the only such SU(2) action on $O(m)$ that covers the sphere action and preserves the connection.

Now suppose that SO(3) acted on $O(m)$ in this way. Composition with the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ would give such an SU(2) action on which $-I$ acts trivially. By uniqueness, that action would be the natural one, where $-I$ acts on every fibre by $(-1)^m$. Hence for odd m no SO(3) action on $O(m)$ that is linear and unitary on fibres covers the rotation action and preserves the connection. This conclusion does not use the orbit normalisation.

This is the sign of Lecture 13 on a fibre. The lift to SU(2) of the rotation through 2π about the north pole is $-I$, and it acts on the fibre by $(-1)^m$. For odd m, the calculation above shows that no change of phases removes this sign. So SO(3) acts only projectively on the sections of odd degree, and its cover SU(2) acts as a representation.

A spin-one comparison with functions on the sphere

Functions on the sphere contain every integer spin, by Lecture 12, and spin one is their linear part. Holomorphic sections of $O(2)$ contain spin one alone. We compare the two copies. For $r = \hbar$, the integer is $m = 2$. The quantisation is the three-dimensional space of homogeneous quadratic polynomials in (v_0, v_1)

$$F(v_0, v_1) = Av_0^2 + Bv_0v_1 + Cv_1^2.$$

There is another spin-one model in Lecture 12: the linear coordinate functions on the sphere. The two models are isomorphic as representations, although a bundle section and a scalar function are different geometric objects.

We now identify the two models explicitly. Write $F(v) = v^T Q v$ and set

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T(Q) = \epsilon^{-1} Q = \begin{pmatrix} -B/2 & -C \\ A & B/2 \end{pmatrix}.$$

This is an invertible linear map from symmetric matrices to traceless matrices. The polynomial action sends Q to $g^{-T} Q g^{-1}$. Since $g^T \epsilon g = \epsilon$ for $g \in \mathrm{SL}(2, \mathbb{C})$,

$$T(g^{-T} Q g^{-1}) = g T(Q) g^{-1}.$$

Thus T intertwines the quadratic section representation with the complexified adjoint representation.

For $Y \in \mathfrak{su}(2)_{\mathbb{C}}$, let ℓ_x be the functional with coordinates $x_j = \ell_x(X_j)$ and define the linear function $f_Y(x) = \ell_x(Y)$ on the sphere, extending ℓ_x complex-linearly. Under pullback by g^{-1},

$$f_Y(g^{-1} \cdot x) = \ell_x(\mathrm{Ad}(g)Y) = f_{\mathrm{Ad}(g)Y}(x).$$

Hence the map sending Y to f_Y is another intertwiner. The three basis elements X_j give the independent functions x_j, so this map is an isomorphism onto their span. Since $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$, the map $Q \mapsto f_{T(Q)}$ identifies the two spin-one models explicitly. The normalised period is still $m = 2$, the dimension is three, and the action descends to $\mathrm{SO}(3)$.

27.5 The compact construction and the shift

This section compares the construction with the recipe of Lecture 19. There a subalgebra $\mathfrak{m}$ with $\ell([\mathfrak{m}, \mathfrak{m}]) = 0$ gave a character χ_ℓ of the group of $\mathfrak{m}$, and we induced from it. In $\mathfrak{su}(2)$ this fails for a nonzero ℓ. A real polarisation would have dimension two, and $\mathfrak{su}(2)$ has no two-dimensional subalgebra, because the bracket of two independent vectors is perpendicular to their plane. So we pass to the complexified algebra and take the positive Borel subalgebra.

The sphere illustrates the general compact picture. Let K be compact, connected, simply connected and semisimple. Let λ be a dominant integral weight in the real convention $d\chi = i\lambda$. Recall the positive Borel subalgebra fixed in Lecture 26. Our sign conventions follow that choice. Extend λ by zero on the root directions, and take the orbit through $\ell = -\hbar\lambda$. By the orbit normalisation, the stabiliser of ℓ acts on the fibre over ℓ by a character with differential $-i\lambda$. Extended by zero on the root directions, ℓ vanishes on $[\mathfrak{b}, \mathfrak{b}]$, which is the sum of the positive root spaces. The character of B with differential $i\ell/\hbar$ is therefore χ_λ^{-1}. So the rule $f(gm) = \chi_\ell(m)^{-1} f(g)$ of Lecture 19 becomes $f(gb) = \chi_\lambda(b) f(g)$, which is the rule for the sections of L_λ in Lecture 26.

For strictly dominant λ, the stabiliser is a maximal torus T. The orbit is K/T and the bundle is L_λ of Lecture 26. Choose its invariant complex structure from the positive Borel subalgebra. The Borel–Weil theorem of Lecture 26 then gives V_λ^*. The sign in $\ell = -\hbar\lambda$ and the dual cancel. Let w_0 be the longest element of the Weyl group, and put $\mu = -w_0\lambda$, the highest weight of V_λ^*. Since w_0 is represented in K, the orbit through $-\hbar\lambda$ contains $\hbar\mu$. So the irreducible representation of highest weight μ belongs to the orbit through $\hbar\mu$, and the dominant integral weights label integral coadjoint orbits. For $\mathrm{SU}(2)$, the weight m has $\lambda(X_3) = -m/2$, so $-\hbar\lambda$ is the north pole of the sphere of radius $\hbar m/2$. For a weight on a chamber wall, the stabiliser K_ℓ is larger. The orbit is then the partial flag manifold K/K_ℓ. The analogous line bundle and holomorphic-section result are cited from Kirillov, Chapter 5, Sections 2–3. We do not prove that general geometric construction here.

For a different global form of K, the stabiliser character must exist for the actual stabiliser group. This is the equivariant integrality test. It reproduces the character-lattice restriction of Lecture 8. An ordinary integral period condition alone does not encode the global form.

Kirillov's character formula, which we do not prove, integrates over the orbit through $\lambda+\rho$ (Kirillov, Chapter 5). On the sphere, $r^2 = j^2\hbar^2$, while the Casimir is $j(j+1)\hbar^2 = \big((j+\frac{1}{2})^2-\frac{1}{4}\big)\hbar^2$. So the shift appears in quantum formulas and not in the orbit.

No ρ shift was used in this construction. On the sphere, the orbit label is m, and the uncorrected section space has dimension $m+1$. The character nevertheless contains $m+1$:

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin((m+1)\theta)}{\sin\theta} \qquad (\sin\theta \neq 0).$$

This is the Weyl character formula, with its familiar $\lambda+\rho$ numerator. It does not say that the prequantum orbit above had label $m+1$. Half-form corrections are another construction and also require a convention. Optional Exercise 27.6 shows one such choice on the sphere, and nothing in this lecture depends on it.

Self-checks

1. If $r = 3\hbar/2$, find the normalised period and the quantised representation.

Answer. The period divided by $2\pi\hbar$ is $2r/\hbar = 3$. The bundle is $\mathcal{O}(3)$ and its sections give V_3, of dimension four and spin $3/2$.

2. Does that sphere admit an SO(3)-equivariant prequantum lift with the orbit normalisation?

Answer. No. Its stabiliser character would be $e^{3it/2}$ on a rotation of period 2π. It takes the value -1 at $t = 2\pi$.

3. Why can the connection have local potentials although no global potential exists?

Answer. The difference of two local potentials is the logarithmic differential of a transition function. A nonzero curvature period prevents a single global primitive but is consistent with these local potentials.

Summary

The SU(2) orbit of radius $r > 0$ has symplectic area $4\pi r$. Prequantisation requires $m = 2r/\hbar$ to be integral. Holomorphic sections of $\mathcal{O}(m)$ give spin $m/2$. Equivariance for SO(3) further requires even m. The ordinary period, the global equivariant lift and the holomorphic selection are separate steps. The orbit through $-\hbar\lambda$ and the dual V_λ^* of Borel–Weil combine to attach V_μ to the orbit through $\hbar\mu$. For SU(2) this attaches V_m to the sphere of radius $\hbar m/2$.

Reading

Kirillov, *Lectures on the Orbit Method*, Chapter 1, Section 2.4, for integrality, and Chapter 5, Sections 2–3, for compact orbits, holomorphic constructions and character formulas. Sepanski, *Compact Lie Groups*, Section 7.4, states the Borel–Weil theorem proved in Lecture 26.